

YUIL' SK, Russian Federation

WMO: 236350

Lat: 63.6776N

Lon: 69.6503E

Elev: 86

StdP: 100.30

Time Zone: 5.00 (E05)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.4	(c) -40.3	(d) -46.8	(e) 0.0	(f) -44.5	(g) -42.9	(h) 0.1	(i) -40.2	(j) 6.7	(k) -15.2	(l) 5.9	(m) -11.8	(n) 0.6	(o) 120	(p) 0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.7	18.8	26.5	17.7	24.2	16.7	20.0	26.1	18.7	24.6	17.5	22.9	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.0	23.0	16.5	11.8	21.5	15.2	10.9	20.1	57.9	25.4	53.6	24.7	49.7	22.9	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.5	(b) 5.5	(c) 4.8	(e) -46.2	(f) 30.8	(g) 3.8	(h) 1.6	(i) -49.0	(j) 32.0	(k) -51.2	(l) 32.9	(m) -53.3	(n) 33.9	(o) -56.1	(p) 35.1		
			DB													
			WB													
				(e) -46.0	(f) 21.0	(g) 3.6	(h) 1.5	(i) -48.6	(j) 22.1	(k) -50.7	(l) 22.9	(m) -52.8	(n) 23.8	(o) -55.4	(p) 24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-23.1	-18.9	-9.5	-4.3	3.6	13.5	17.2	12.7	6.4	-2.4	-14.2	-18.8
	DBStd	15.06	9.57	9.07	7.70	6.23	5.55	5.31	4.69	3.71	3.81	5.47	8.64	10.20
	HDD10.0	5252	1027	809	604	428	215	27	3	16	120	385	726	893
	HDD18.3	7871	1285	1042	862	678	457	161	79	178	358	643	976	1152
	CDD10.0	486	0	0	0	0	17	131	227	99	12	0	0	0
	CDD18.3	64	0	0	0	0	1	14	44	4	0	0	0	0
	CDH23.3	666	0	0	0	0	19	135	476	36	0	0	0	0
	CDH26.7	174	0	0	0	0	2	23	146	4	0	0	0	0
Wind	WSAvg	2.3	1.9	1.9	2.6	2.7	2.6	2.5	2.1	2.1	2.2	2.3	2.1	2.0
Precipitation	PrecAvg	765	48	42	55	61	58	81	75	79	62	81	69	54
	PrecMax	943	84	72	111	103	98	126	213	170	103	134	104	94
	PrecMin	643	19	12	9	10	24	42	13	40	23	24	29	28
	PrecStd	69	17	15	22	21	20	20	40	25	22	22	19	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	0.0	5.6	10.5	26.0	29.1	31.7	27.0	19.3	11.1	3.3	-0.8
		MCWB	-2.2	-1.2	1.6	4.7	15.7	18.9	20.9	18.8	13.3	8.8	2.8	-1.6
	2%	DB	-4.8	-1.9	3.4	8.1	21.6	26.7	30.1	23.9	16.4	8.5	0.3	-2.0
		MCWB	-5.4	-2.6	0.1	3.4	13.6	18.0	19.8	17.1	11.9	7.2	-0.2	-2.6
	5%	DB	-7.3	-3.9	1.9	6.1	17.2	24.7	28.2	21.7	14.2	6.5	-1.1	-3.7
		MCWB	-7.9	-4.7	-0.6	3.2	11.6	16.7	18.5	15.3	10.8	4.9	-1.7	-4.3
	10%	DB	-9.9	-6.4	0.6	4.3	12.9	22.3	26.0	19.8	12.4	4.2	-2.5	-5.9
		MCWB	-10.4	-7.1	-1.2	1.8	7.9	15.5	17.6	14.6	9.9	3.0	-3.0	-6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.1	-1.1	2.0	5.9	16.4	20.7	22.2	19.8	13.9	8.9	2.6	-1.4
		MCDB	-1.3	-0.4	4.5	7.9	24.7	27.1	29.5	25.0	18.0	10.7	3.1	-0.7
	2%	WB	-5.3	-2.6	0.5	4.2	14.0	19.0	20.9	17.8	12.5	7.4	-0.1	-2.6
		MCDB	-4.8	-1.9	2.5	7.0	20.7	25.1	27.5	22.2	15.7	8.7	0.3	-2.0
	5%	WB	-7.9	-4.6	-0.3	3.2	11.6	17.5	19.6	16.4	11.2	5.2	-1.5	-4.2
		MCDB	-7.3	-3.9	1.7	5.8	17.1	23.0	25.6	20.6	13.5	6.3	-1.2	-3.7
	10%	WB	-10.3	-7.1	-1.2	2.1	8.8	16.2	18.5	15.2	10.2	3.0	-3.0	-6.4
		MCDB	-9.9	-6.4	0.6	4.4	12.5	21.1	24.3	18.6	12.2	3.9	-2.4	-5.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.3	10.4	10.3	9.2	9.7	10.4	8.9	6.7	5.0	8.0	7.9
		MCDBR	10.3	8.6	8.9	10.5	14.7	12.7	13.7	13.0	9.7	6.6	7.5	5.7
		MCWBR	10.3	7.8	6.5	7.3	8.1	5.7	5.1	6.9	5.8	5.0	7.4	5.4
	5% WB	MCDBR	10.4	7.9	7.8	9.3	14.0	11.7	11.9	11.0	8.6	5.5	7.4	6.0
		MCWBR	10.3	7.8	6.0	6.8	7.7	5.9	5.4	6.4	5.6	4.8	7.4	5.7
Clear-Sky Solar Irradiance	taub	0.225	0.280	0.329	0.351	0.368	0.384	0.415	0.388	0.339	0.303	0.225	0.224	
	taud	1.915	2.034	2.009	2.080	2.296	2.374	2.282	2.334	2.435	2.283	1.960	1.894	
	Ebn at Noon	338	615	732	813	841	834	790	778	752	605	331	128	
	Edh at Noon	32	69	108	126	112	106	114	99	73	54	28	15	
All-Sky Solar Radiation	RadAvg	0.21	1.01	2.60	4.39	5.17	5.16	5.02	3.42	1.99	0.84	0.26	0.08	
	RadStd	0.01	0.08	0.15	0.45	0.46	0.37	0.57	0.36	0.21	0.10	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page